

Def. Let M be an oriented regular surface. A map

$$V: M \rightarrow S^2; p \mapsto n(p)$$

is called the Gauss map.

Since $V(p)$ is perpendicular to T_p and also perpendicular to $T_p S^2$, therefore $T_p M$ and $T_p S^2$ are parallel.

Let $x \in T_p M$. Choose a smooth curve

$$\alpha: (-\varepsilon, \varepsilon) \rightarrow M \text{ such that } \alpha(0) = p \text{ and } \alpha'(0) = x.$$

Now, define $dN_p(x) = \left. \frac{d}{dt} \right|_{t=0} V(\alpha(t))$

As usual convention, define the shape operator is

$$S_p: T_p M \rightarrow T_p M; x \mapsto -dN_p(x).$$

Consider a patch $X: U \rightarrow M$ s.t. $X(u_0, v_0) = p$, and $x = aX_u + bX_v$.

$$dN_p(x) = aN_u + bN_v.$$